

535/1

PHYSICS

Paper 1

Nov/Dec 2025

1 hour 30 Minutes

UGANDA CERTIFICATE OF EDUCATION

S.2 END OF YEAR ASSESSMENT

PHYSICS

PAPER 1

(THEORY)

1 HOUR 30 Minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of a total of **three** items.

Answer only **three** items.

You may use appropriate illustrations in your responses.

Graph papers are provided.

@2025 PHY DEPARTMENT

Item 1

A child observes images formed when he looks at the two curved surfaces of a spoon. He realizes that the inner surface provides an image that is upside down whereas the outer surface produces an upright image. The child does not understand why the two images appear different, and approaches you for assistance.

Task

- (a) (i) Using relevant diagrams, explain to the child how each surface of the spoon forms an image.
- (ii) For each of the surfaces of the spoon, state other two properties of the image formed
- (b) A student places an object at the centre of curvature of a concave mirror of focal length 10.0 cm. A second student places an object at a distance of 8.0 cm from the same mirror. Compare the images obtained by the two students.
- (c) State and explain two applications of spoon-like shaped mirrors.

Item 2.

On a construction site, there is always a mechanical test done on all construction materials before they are accepted or rejected. The following table shows the different materials and other details. A material can only be accepted if the Young's modulus exceeds $2 \times 10^7 \text{ Nm}^{-2}$.

Item	Quantity	Price per unit	Stress to be undergone (Pa)	Strain to be experienced
Circular metallic sheets	40 pieces	UGX 60,000	4000	0.00015
Nails	50 kg	UGX 7000	6000	0.0006

You have been approached by a young man who wants to construct his first house in life and has no experience in construction materials.

Task: Using your knowledge of physics, help the young man to;

- (a) The meaning of the words stress, strain and young's modulus
- (b) Know which material would be accepted and the amount of money that is used in the purchase.
- (c) Determine the amount of money that was wasted in the purchase if the item was rejected.

- (d) Additionally, the young man is recommended to use **steel bars (hollow), damp proof course, reinforced concrete** for the foundation of a house.

Explain to the young man the;

- I. Meaning of reinforced concrete, suggest materials used in reinforcement.
- II. Why damp proof course is important in the work?
- III. Why steel bars are necessary?

Item 3.

A business man owns a salon car and uses it for special hire. One day, while on duty his car developed a mechanical problem arising from the braking system of the car. He took the car to the garage for servicing.

The car was lifted using a hydraulic lift whose pistons had areas as 0.03 m^2 and 0.02 m^2 respectively. The smaller piston of the hydraulic press was pushed down with a force of 120 N and the car, placed on a larger piston was seen rising up for clear checkup. The mechanic also discovered that the same car had broken side mirrors that needed replacement.

Task.

As a learner of physics

- (a) Name the suitable mirror that should be bought to replace the broken mirrors
- (b) Draw ray diagrams to show and explain why such a mirror is preferred to others for this purpose.
- (c) State the principle on which hydraulic lift works.
- (d) What force was required to push a larger piston so as to raise the car?